# REQUIREMENT ANALYSIS

## Data Flow Diagram & User Stories

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| **Date** | 11 October 2026 |
| **Team ID** | SWTID-2026-7837 |
| **Project Name** | Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer |
| **Module** | ServiceNow Service Catalog / Requested Item / Catalog Task |

## 1. Purpose

This document defines the data flow and user interactions for the Standard Laptop Procurement Automation project. The design shows how a Standard Laptop request moves from the Service Catalog through approval and how Flow Designer automatically creates and assigns a Catalog Task to the Hardware team.

## 2. System Scope

The data flow covers the Service Catalog request, Requested Item, approval outcome, Flow Designer automation, Catalog Task creation, and assignment of the resulting task to the Hardware team for laptop configuration.

## 3. Data Flow Diagram – Context / Level 0

| IT Procurement User | ServiceNow Service Catalog | Approval Process | Flow Designer | Catalog Task / Hardware Team |
|---|---|---|---|---|
| Submits Standard Laptop request | Creates Requested Item | Request is reviewed and approved | Detects approved request and creates task | Receives assigned task for configuration |

Level 0 interpretation: The user submits a Standard Laptop request through the Service Catalog. After approval, Flow Designer processes the approved request, creates the required Catalog Task, and assigns it to the Hardware team. The result is an actionable task without manual task creation or assignment.

## 4. Data Flow Diagram – Level 1

| Step | Process | Input / Condition | Output |
|---|---|---|---|
| 1 | Receive Standard Laptop request | Service Catalog / Requested Item | Request context available for processing |
| 2 | Evaluate approval outcome | Approval status of the request | Approved request proceeds to automation |
| 3 | Start Flow Designer workflow | Approved Standard Laptop request | Configured flow actions execute |
| 4 | Create Catalog Task | Approved request details | Catalog Task created |
| 5 | Assign Hardware team | New Catalog Task | Task routed to Hardware team |
| 6 | Begin configuration | Assigned Catalog Task | Hardware team has actionable work item |

Key data movements: the Standard Laptop request provides the Requested Item context; approval status determines whether the automation proceeds; Flow Designer creates the Catalog Task; and the task is assigned to the Hardware team for configuration.

## 5. User Stories

| ID | User Story | Acceptance Criteria |
|---|---|---|
| US-01 | As an IT procurement requester, I want my Standard Laptop request to move automatically after approval so that I do not need to coordinate task creation manually. | When the Standard Laptop request is approved, the configured automation starts processing the request. |
| US-02 | As an IT procurement requester, I want a Catalog Task to be created automatically after approval so that the approved request continues to the next procurement step. | An approved Standard Laptop request results in a Catalog Task being created. |
| US-03 | As a Hardware team member, I want the Catalog Task to be assigned to the Hardware team so that the correct team can begin laptop configuration. | The generated Catalog Task is assigned to the Hardware team. |
| US-04 | As a Hardware team member, I want the approved request information to be available through the Catalog Task so that I can act on the procurement request. | The Catalog Task is created from the approved Standard Laptop request and provides the required work context. |

| ID | User Story | Acceptance Criteria |
|---|---|---|
| US-05 | As a procurement process owner, I want approved Standard Laptop requests to follow a consistent workflow so that manual errors and delays are minimized. | The defined Flow Designer process consistently creates and assigns the Catalog Task for approved requests. |
| US-06 | As a procurement process owner, I want task creation and assignment to be automated so that task visibility improves and configuration can begin promptly. | No manual Catalog Task creation or assignment is required for the automated path. |

## 6. User Interaction Flow

1. The user submits a Standard Laptop request through the Service Catalog.
2. ServiceNow creates the corresponding Requested Item.
3. The request proceeds through the approval process.
4. When the request is approved, Flow Designer starts the configured automation.
5. The flow creates a Catalog Task for the approved Standard Laptop request.
6. The flow assigns the Catalog Task to the Hardware team.
7. The Hardware team receives the actionable task and can begin laptop configuration.

## 7. Data Elements Involved

| Data Element | Role in Data Flow | Control / Process |
|---|---|---|
| Standard Laptop Request | Represents the user's procurement request. | Service Catalog |
| Requested Item | Carries the requested item context into the procurement workflow. | Service Catalog / Requested Item |
| Approval Status | Determines whether the automation should proceed. | Approval Process |
| Flow Designer Flow | Coordinates the automated approval-to-task transition. | Flow Designer |
| Catalog Task | Represents the actionable hardware configuration work. | Catalog Task |
| Hardware Team | Receives the generated task for configuration. | Task Assignment |

## 8. Traceability to the Project Problem

The data flow directly addresses the problem of manual task creation and assignment in Standard Laptop procurement. The flow connects the Service Catalog and approval process to Flow Designer, which automatically creates and assigns the Catalog Task. This reduces manual effort, improves visibility, minimizes errors, and enables the Hardware team to begin configuration promptly.

## 9. Summary

The proposed data flow provides a simple end-to-end automated procurement workflow. A Standard Laptop request enters through the Service Catalog, passes through approval, and then triggers Flow Designer to create and assign a Catalog Task to the Hardware team. The user stories define the expected behavior in a clear and testable form and maintain direct traceability to the project problem and solution objective.